

EDUCATION

Stanford University

Autumn 2025 to Winter 2027

Master of Science, Mechanics and Computation

Stanford, CA, USA

• **Relevant Coursework:** Solid Mechanics, Linear Algebra with applications to Engineering Computations, Advanced Structural Analysis

Birla Institute of Technology & Science, Pilani (BITS Pilani)

Aug 2021 - May 2025

Bachelor of Engineering, Civil Engineering — **CGPA: 9.26/10**

Pilani, India

• **Relevant Coursework:** Mechanics of Solids, Analysis of Structures, Design of Reinforced Concrete Structures, Foundation Engineering, Design of Steel Structures, Earthquake Resistant Design and Construction, Introduction to FEM

TECHNICAL SKILLS

Software/ Tools: DIANA FEA, ABAQUS, ETABS, MASTAN2, AutoCAD, Revit, MS Office Suite, Git, Github

Programming: C++, MATLAB, Python, Julia, C, $\LaTeX$

PUBLICATIONS

- *Experimental and numerical studies on reinforced concrete base restrained walls and tanks subjected to shrinkage effects* Amrit Bahl, Mohammad Najeeb Shariff, and Sankati Yellamanda
Accepted in the *ACI Structural Journal*, Preprint available: International Concrete Abstracts Portal
- *A comparison of analytical models for crack-width prediction in RC ties subject to direct tension and restrained shrinkage* A R Bahl, Pawan Prakash, and M N Shariff
Accepted in the *4th R.N. Raikar Memorial International Conference 2024*
- *Dynamic stability of randomly distributed CNT reinforced composite beam using the generalized differential quadrature method* A R Bahl, and Rajesh Kumar
Accepted as a book chapter in the book *Experimental and Mathematical Modeling of Aerospace and Civil Engineering Composite Structures* by Apple Academic Press (CRC Press/ Taylor and Francis)

RESEARCH EXPERIENCE

Design optimization of composite compressed air energy storage (CAES) pressure vessel Jun 2025 - Sep 2025

Project Associate: Mentor: Prof. Rajesh Kumar, BITS Pilani — Sponsored by: Siemens Energy

- Validated the burst pressure and stress-strain behavior of CFRP and GFRP-Steel composite pressure vessels with experimental results using a progressive failure based non-linear finite element analysis in ABAQUS
- Conducted a parametric study on the designed prototype with joints and holes to find the optimum fiber winding angle that reduced the total material used to sustain a required design pressure of 110 bar

Numerical study on RC base-restrained walls and tanks subjected to shrinkage effects Jun 2024 - Dec 2024

Undergraduate Thesis: Mentor: Prof. Mohammad Najeeb Shariff, Indian Institute of Technology Bombay

- Compared existing analytical models in direct tension and restrained shrinkage to determine the accuracy and effect of the various parameters on crack-width using Sobol's sensitivity analysis
- Validated the crack-widths, crack-patterns, and shrinkage strains from the experimental tests done on base-restrained walls and tanks using a 3D nonlinear finite element (FE) analysis on DIANA FEA
- Fitted an equation to find the variation of the restraint factor along the height of a wall for different aspect ratios derived using a linear FE analysis to assess the tensile stresses due to a base-restraint

Dynamic stability analysis of randomly distributed CNT reinforced composite beam Jan 2024 - May 2024

Research Assistant: Mentor: Prof. Rajesh Kumar, BITS Pilani

- Modeled the effective three-phase CNT-Fiber-Composite system using the Eschelby-Mori-Tanaka scheme and the Chamis homogenization technique
- Solved the governing differential equations for the First-Order Shear Deformation theory using the Generalized Differential Quadrature method to validate the natural frequency and critical buckling load
- Conducted a parametric study to find the effect of agglomeration ratio on the instability regions using Bolotin's method

Susceptibility of soil-rock mixture slopes to Landslides

Aug 2023 - Dec 2023

Research Assistant: Prof. Nishant Roy, BITS Pilani

- Reviewed the existing literature to conduct a bibliometric analysis on modeling slope stability of soil-rock mixtures
- Developed a MATLAB code to generate randomly distributed circular and spherical blocks in an embankment configuration following a gradation of rock particles
- Exported the complicated geometries generated in MATLAB to Optum G2 and Optum G3 through the software's backend to create 2500 models with varying parameters

PROJECTS

User defined 3D elastic first-order analysis program for MASTAN2

Oct 2025 - Dec 2025

Course Project: *Advanced Structural Analysis* | Prof. Gregory G Deierlein, Stanford

- Developed a program to run any user-defined 3D elastic first-order analysis in MASTAN2 using Object Oriented Programming in MATLAB
- The program efficiently calculated the element forces, nodal displacements, support reactions and the error between the applied and back-calculated forces

Design and Analysis of a Multi-Storey Steel Shopping Mall

Jan 2024 - May 2024

Course Project: *Design of Steel Structures* | Prof. Shamsher Bahadur Singh, BITS-Pilani

- Designed and analyzed a G+3 steel shopping mall structure in ETABS using the Limit State method as per IS 800:2007
- Checked the design of critical members using hand calculations as per IS 800:2007

Design of a standalone apartment

Jan 2023 - May 2023

Course Project: *Construction Planning and Technology* | Prof. G Muthukumar, BITS-Pilani

- Drafted the plan of a G+0 residential apartment on AutoCAD and modeled the 3D structure with plumbing on Revit based on the National Building Code of India 2016
- Conducted a cost and time analysis of the apartment construction to determine the required resources using Primavera P6 and prepared a schedule for the construction activities

TEACHING ASSISTANTSHIPS

Soil Mechanics

Jan 2024 - May 2024

CE F243: Prof. Sayantan Chakraborty, BITS-Pilani

- Explained the 1D consolidation test, through weekly demonstrations, to a group of 50 undergraduate students
- Responsible for the invigilation of evaluative components and the doubt-clearing sessions of around 90 undergraduate students enrolled in the course

Fluid Mechanics

Aug 2023 - Dec 2023

CE F231: Prof. Selva Balaji, BITS-Pilani

- Prepared questions and their solutions on each of the 10 modules covered in the course
- Responsible for the invigilation of evaluative components and the doubt-clearing sessions of around 90 undergraduate students enrolled in the course

LEADERSHIP AND EXTRA CURRICULARS

Student Alumni Relations Head | Head of Events | BITS-Pilani

- Led a team of 25 members to organize the batch photograph and the Director's Tea Party for the graduating batch of around 900 students
- Coordinated and organized the reunion of 3 alumni batches hosting around 100 alumni from each batch

Hindi Press Club | Senior Editor | BITS-Pilani

- Served as a writer and then the editor of the monthly magazines and the special issues published by the Hindi Press Club of BITS Pilani circulated among 4000+ students and faculty members
- Worked in a team of 30 members to ensure the coverage of all events during the cultural, technical, and sports festival at BITS by publishing daily newsletters read by students and participants

Student-Faculty Committee | Student Representative | BITS-Pilani

- Selected as one of the 5 representatives from a batch of 90 students to address the academic and non-academic issues of the student body and find effective solutions through bi-semesterly meetings
- Presented the students' standpoint on potential electives, workshops, and activities conducted in the department

National Service Scheme | Volunteer | BITS-Pilani

- Conducted bi-weekly lectures on English and Personality Development for underprivileged children from classes 6th to 12th
- Organized the Old Notebook Collection Drive with 5 team members by coordinating with 100+ volunteers and collected 200 kilograms of scrap paper

ACADEMIC ACHIEVEMENTS

Poster Presentation Competition - Young Researchers Conclave

- Awarded the second position in the poster presentation competition organized as a part of the Young Researchers Conclave at the CSIR-CRRI for the poster titled *Stabilization of Black Cotton Soil with Alkali Bypass dust*